9 A beam of white light is dispersed in a glass prism. Figure 9.1 shows the paths taken by green, indigo, red and yellow light.

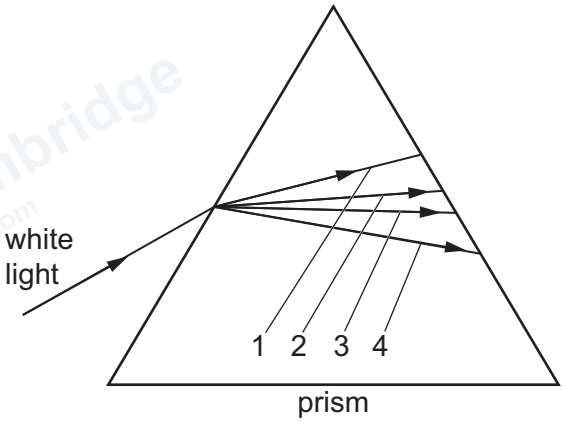

Figure 9.1

(a) (i) State which of the rays, 1, 2, 3 or 4, is:

- the green ray .....
- the red ray. .....

[2]

(ii) The light leaves the prism.

On Figure 9.1, continue ray 1 and ray 4 to show the path of light in the air. [2]

(b) (i) The Sun emits most of its energy in three regions of the electromagnetic spectrum. Visible light is one region.

State the other **two** regions.

1 .....

2 .....

[1]

(ii) The Sun is a star in the Milky Way galaxy.

State the approximate diameter of the Milky Way. Give your answer in light-years.

diameter = ... light-years [1]

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(c) Scientists on the Earth observe light emitted by a distant galaxy. The observed light is redshifted.

(i) State what is meant by redshifted.

.....

..... [1]

(ii) Explain how redshift supports the Big Bang Theory.

.....

.....

..... [2]

[Total: 9]

Question 10 is on the back page.

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